Analyzing Drug–Gene Interactions in Cancer Therapy and Drug Metabolism

Abstract

This study analyzes how cancer drugs interact with human genes and how these drugs are metabolized in the body. Using data from the Drug-Gene Interaction Database (DGIdb), the project identifies the most commonly targeted genes by FDA-approved cancer drugs and examines interactions with CYP450 enzymes. Results from this study show that a small number of genes, such as EGFR, are frequently targeted, while enzymes like CYP3A4 play a major role in drug metabolism. These findings highlight the importance of both gene targeting and metabolism in treatment effectiveness. This study also provides insight into how cancer therapies function and supports the role of personalized medicine.

Introduction

Cancer treatments often depend on targeting specific genes involved in tumor growth while also considering how drugs are processed in the body. These drug–gene interactions are important for understanding how treatments work and why some therapies are more effective than others. Drug metabolism also plays a major role in treatment outcomes, especially through the cytochrome P450 (CYP450) family of enzymes, which metabolize many cancer drugs. Since both gene targeting and metabolism influence patient response, this project focuses on identifying the genes most commonly targeted by FDA-approved cancer drugs and determining which drugs interact with CYP450 enzymes. By examining these patterns, the study aims to understand the relationship among gene targeting, drug metabolism, and personalized medicine.

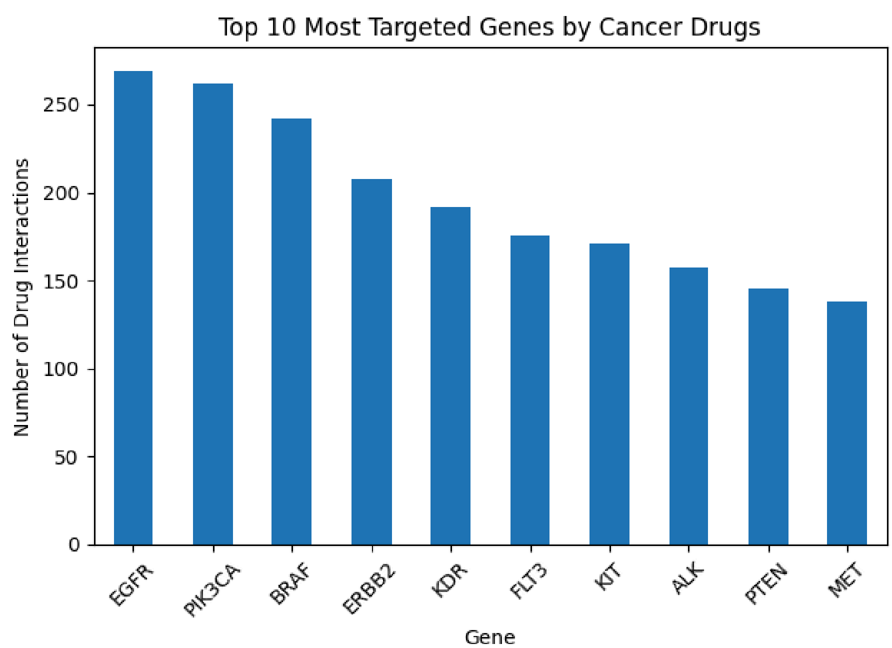
Methods

The data used in this project were taken from the Drug-Gene Interaction Database (DGIdb). This dataset was uploaded to Google Colab and analyzed using Python and the pandas library. First, the data was filtered to include only cancer-related drugs using the anti_neoplastic column. Then, gene targets were counted to find which genes appeared most often. To study drug metabolism, the data were further filtered to include only CYP450 genes, which represent enzymes that process drugs. Bar charts were created using matplotlib to show patterns in the data. A table was also made to list examples of drugs associated with each CYP450 enzyme.

Results

Figure 1: Top 10 Most Frequently Targeted Genes by Cancer Drugs

This bar chart displays the ten genes most commonly targeted by FDA-approved cancer drugs.

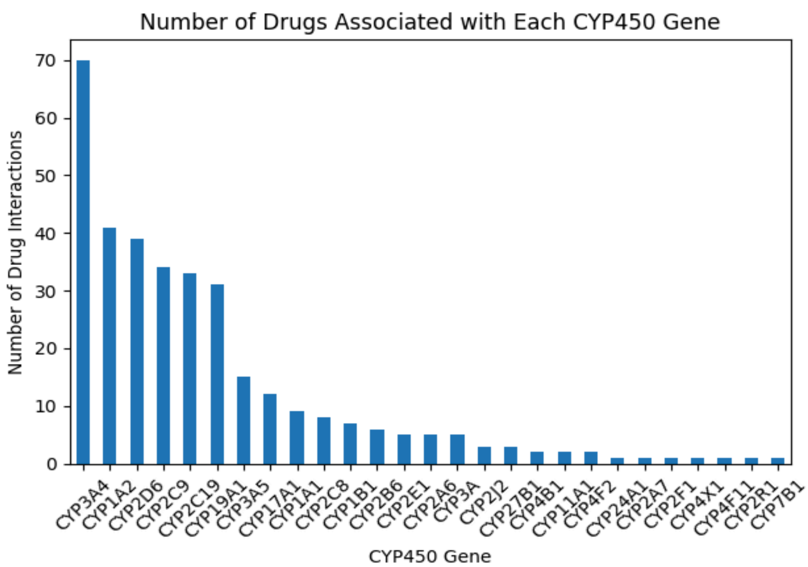


This figure shows the top 10 genes most frequently targeted by cancer drugs. These genes likely play a major role in cancer growth or survival, making them important treatment targets.

The EGFR gene (Epidermal Growth Factor Receptor) is the most frequently targeted, which aligns with its role in regulating cell growth; mutations can keep it active and promote uncontrolled cell division. The concentration of drugs around a small group of genes suggests that many therapies focus on key biological pathways. Overall, the chart highlights patterns in how cancer treatments are designed and targeted.

Figure 2: Number of Drugs Associated with Each CYP450 Gene

This bar chart shows how many cancer drugs interact with each CYP450 enzyme.



This figure shows that some CYP450 enzymes metabolize more cancer drugs than others. CYP3A4 appears most frequently because it processes many medications, including cancer therapies, while enzymes like CYP7B1 are less common because they are more involved in cholesterol metabolism than drug metabolism (MedlinePlus, 2023). The uneven distribution suggests that certain enzymes play a larger role in drug breakdown and interactions, which may affect side effects and patient response.

Table 1: Table of CYP450 Enzymes Interacting with Cancer Drugs

This table lists CYP450 enzymes and the top 5 different cancer drugs the enzymes interact with

CYP450 Gene	Top 5 Associated Drugs
CYP3A4	ALECTINIB, AMSACRINE, ASPIRIN, AZATHIOPRINE, CABAZITAXEL
CYP1A2	AMSACRINE, ASPIRIN, AZATHIOPRINE, CAMPTOTHECIN, CARMUSTINE
CYP2D6	AMSACRINE, ASPIRIN, AZATHIOPRINE, CARMUSTINE, CHLOROQUINE
CYP2C9	AMSACRINE, ASPIRIN, AZATHIOPRINE, CARMUSTINE, CELECOXIB
CYP2C19	AMSACRINE, ASPIRIN, AZATHIOPRINE, BOSUTINIB, CAMPTOTHECIN
CYP19A1	ALENDRONATE SODIUM, AMINOGLUTETHIMIDE, ANASTROZOLE, CAPECITABINE, EXEMESTANE
CYP3A5	CABAZITAXEL, CARBOPLATIN, CELECOXIB, ERLOTINIB, GEFITINIB
CYP17A1	ABIRATERONE ACETATE, ENZALUTAMIDE, ORTERONEL
CYP1A1	CAPECITABINE, DACARBAZINE, DAUNORUBICIN LIPOSOMAL, DOCETAXEL ANHYDROUS, DOXORUBICIN HYDROCHLORIDE
CYP2C8	CABAZITAXEL, FLUOROURACIL, GEMCITABINE, IBUPROFEN, SODIUM SALT, PAZOPANIB
CYP1B1	AMINOGLUTETHIMIDE, ANASTROZOLE, DOXORUBICIN HYDROCHLORIDE, EXEMESTANE, FLUOROURACIL
CYP2B6	BICALUTAMIDE, IFOSFAMIDE, IMATINIB, METHOTREXATE, SORAFENIB
CYP2E1	CYTARABINE, FLUDARABINE, FLUOROURACIL, GEMTUZUMAB OZOGAMICIN, IDARUBICIN
CYP2A6	DOCETAXEL ANHYDROUS, FLUOROURACIL, IFOSFAMIDE, LETROZOLE, VALPROIC ACID
CYP3A	DOXORUBICIN HYDROCHLORIDE, FLUOROURACIL, LENVATINIB, PACLITAXEL, VEMURAFENIB
CYP2J2	DANAZOL, DAUNORUBICIN LIPOSOMAL, DOXORUBICIN HYDROCHLORIDE
CYP27B1	DECITABINE, PEGINTERFERON ALFA-2A, TRICHOSTATIN A
CYP4B1	DOCETAXEL ANHYDROUS, THALIDOMIDE
CYP11A1	AMINOGLUTETHIMIDE
CYP4F2	ASPIRIN, LOVASTATIN
CYP24A1	ASPIRIN
CYP2A7	MERCAPTOPURINE
CYP2F1	IMATINIB
CYP4X1	FLUOROURACIL
CYP4F11	ASPIRIN
CYP2R1	PEGINTERFERON ALFA-2B
CYP7B1	MITOMYCIN

This figure provides a more detailed view of the top 5 specific drug-gene interactions for CYP450 enzymes, as using the top 5 drugs helps with readability. It allows for a better understanding of some of the more frequent drugs that are interacting with each enzyme. This is useful for understanding how different drugs behave in the body while also showing other possible drug interactions when multiple medications are used. Overall, the trends presented in this table help better support patterns that have been shown in Table 1.

Discussion

The results show that cancer drugs are not spread evenly across all genes. Instead, they focus on a smaller group of important genes, which are likely key to how cancer develops. At the same time, drug metabolism also plays a major role, since many of these drugs interact with CYP450 enzymes. Enzymes like CYP3A4 appear frequently, showing that they are important in processing medications. These patterns help explain why different patients may respond differently to the same treatment. Overall, both gene targeting and metabolism are important when studying how cancer drugs work.

Conclusion

In conclusion, this project shows that cancer treatment depends on both targeting the right genes and how the body processes drugs. A few genes are targeted much more often, and certain enzymes are responsible for handling many drugs. Understanding these patterns can help improve how treatments are developed and used. It also supports the idea of personalized medicine, where treatments are adjusted based on the patient.

References

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